# Bihar Engineering University, Patna
## End Semester Examination - 2023

*Course: B. Tech* | *Semester: VI* | *Time: 03 Hours*

*Code: 103601* | *Subject: Power System – II (Operation and Control)* | *Full Marks: 70*

---

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*

*(ii) There are NINE questions in this paper.*

*(iii) Attempt FIVE questions in all.*

*(iv) Question No. 1 is compulsory.*

---

**Q.1** ***Answer any seven question only:*** **[2 x 7 = 14]**

(a) What is a disturbance and what are the two types of disturbances?

(b) When is a power system said to be steady-state stable?

(c) Why power flow analysis is made?

(d) What are the information that are obtained from a load flow study?

(e) What are the different types of buses in a power system?

(f) Define negative sequence and zero sequence components.

(g) What is meant by a fault?

(h) Write the difference between SVC and STATCOM.

(i) What are the methods of neutral grounding?

(j) What is the function of phase shifters?

**Q.2** (a) Give reasons: **[4]**

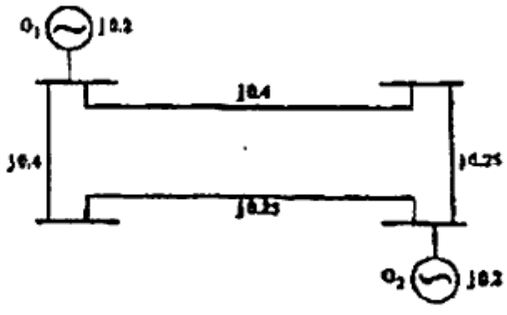
(1) One of the buses is taken as slack bus in load flow studies.

(2) An acceleration factor is commonly used in load flow studies using GS method.

(b) Discuss the Flow chart of load flow solution using Newton-Raphson method for all type of buses. **[10]**

**Q.3** (a) Give comparison of GS method and NR method used for load flow studies of power systems. **[6]**

(b) Form the YBUS by using singular transformation for the network shown below. Including the generator buses **[8]**




**Q.4** (a) With the help of neat diagram explain turbine speed governing mechanism and also derive a mathematical model of turbine speed governing system. **[8]**

(b) Discuss in detail about Automatic Voltage Regulator. **[6]**

HIGHLY CONFIDENTIAL DOCUMENT EDC - 151, @ 04-12-2023 13:08:05
© Apna BEU



Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.5 (a) What is power system stability? Define: Steady State, Transient, and Dynamic Stability. [6]

(b) Drive per unit swing equation of a synchronous machine. Show how machines swinging coherently can be reduced to single machine. [8]

Q.6 (a) Explain equal area criteria of stability. [7]

(b) What is critical clearing angle? Explain it. [7]

Q.7 (a) Explain how state estimation used in power system. [7]

(b) What is a Phasor measurement unit (PMU) in electric power system? [7]

Q.8 (a) Discuss the principle of symmetrical components. Derive the necessary equations to convert: (i) Phase quantities into symmetrical components. (ii) Symmetrical components into phase quantities. [7]

(b) Derive an expression for the fault current for the LG fault. [7]

Q.9 Write short notes on *any two* of the following: [7x2=14]

(i) SCADA System
(ii) Sequence network on transformer
(iii) Power angle Curve
(iv) Formation of Bus Admittance matrix

© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play